

Ayush Bajaj

Software Developer

Github: github.com/AyushBajaj7

Email: ayushbajaj215@gmail.com

Mobile: +91-6205223800

Portfolio: [Portfolio Website](#)

EDUCATION

- **VIT University, Vellore** 2023 – 2027 (Expected)
Bachelor of Technology CGPA: 8.80 / 10
- **DAV SKVB Higher Secondary School**
Class X: 95.6% — Class XII: 80.6% 2019 – 2022

SKILLS SUMMARY

- Programming Languages: Python, Java, C, HTML, CSS
- Web Technologies: React.js, Node.js, Express.js, REST APIs
- Machine Learning: Classification Models, Feature Engineering, Model Evaluation, Transformers
- Cloud & Databases: AWS (EC2, Amazon Polly, IAM), MySQL, MongoDB
- Core Computer Science: Data Structures and Algorithms, OOPs, DBMS
- Tools & Soft Skills: GitHub, VS Code, FFmpeg, Problem Solving, Critical Thinking, Time Management, Adaptability

PROJECTS AND TECHNICAL EXPERIENCE

- **AgriConnect**
Full Stack Developer
 - Designed and developed a full-stack web platform providing real-time agricultural market prices, tools, and government schemes, improving information accessibility for farmers.
 - Built a responsive frontend using React.js and Tailwind CSS to enhance usability and accessibility.
 - Implemented RESTful backend services using Node.js, Express.js, and MongoDB for scalable data management.
 - Centralized critical agricultural information to support data-driven decision-making for farmers.
- **AI-Based PPT Explainer & Video Generator**
Cloud & NLP Developer
 - Developed an AI-driven system to automatically convert presentation slides into narrated explanatory videos.
 - Deployed the end-to-end pipeline on AWS EC2 to perform NLP inference, speech synthesis, and video rendering.
 - Utilized Transformer-based NLP models (FLAN-T5) for slide explanation and Amazon Polly for text-to-speech.
 - Integrated FFmpeg, Boto3, and Hugging Face APIs to automate media processing and workflow execution.
- **AI Powered Disease Prediction Model**
Machine Learning Engineer
 - Built a disease prediction system achieving 95% accuracy through supervised learning and model evaluation.
 - Trained and compared classification algorithms including Extra Tree, Naive Bayes, KNN, Decision Tree, and Neural Networks.
 - Improved early disease diagnosis through feature engineering and performance optimization.
- **3D Architecture Visualization System**
Graphics & Visualization Developer
 - Designed an interactive 3D visualization system for realistic architectural walkthroughs and design validation.
 - Developed 3D modeling and rendering workflows using Blender and Unity.
 - Integrated Python and OpenGL components for dynamic visualization and immersive interaction.

ACHIEVEMENTS

- Caterpillar CODE-A-THON 2024: Collaborated in a competitive hackathon to design solutions for real-world engineering problems.
- DevJams'24 – Google Developer Student Clubs: Participated in rapid prototyping and collaborative software development.
- Yantra Central Hack – IEEE Computer Society: Worked on innovative technical solutions under strict time constraints.
- CookOff 9.0 – CodeChef: Competed in a competitive programming contest, improving algorithmic problem-solving skills.